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PROJECT REPORT ENTITLED
GOVERNANCE, RISK AND COMPLIANCE AUTOMATION WITH
LARGE LANGUAGE MODELS

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DECLARATION

We declare that this project work is our own work. It is being submitted for the degree of Bachelor of Science in Computer Science and Engineering in the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted for any degree or examination in any other University

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DEDICATION

We dedicate this project to our families for their unwavering support and encouragement throughout our academic journey. To our parents, who have sacrificed so much to provide us with the opportunity to pursue our dreams, this achievement is as much yours as it is ours. We also dedicate this work to our colleagues and friends at UMaT for the shared moments of learning and growth that made this project possible.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
AML	Anti-Money Laundering
API	Application Programming Interface
BYOK	Bring Your Own Key
EU AI Act	European Union Artificial Intelligence Act
GDPR	General Data Protection Regulation
GRC	Governance, Risk and Compliance
GPU	Graphics Processing Unit
LIME	Local Interpretable Model-agnostic Explanations
LLM	Large Language Model
LoRA	Low-Rank Adaptation
NIST	National Institute of Standards and Technology
RAG	Retrieval-Augmented Generation
RegTech	Regulatory Technology
SHAP	SHapley Additive exPlanations
SWIFT	Society for Worldwide Interbank Financial Telecommunication
TPM	Trusted Platform Module
XAI	eXplainable Artificial Intelligence

CHAPTER 1

INTRODUCTION

1.1 Background

Banking sector is a sector that is at a critical crossroad of current explosive technological change and heightened regulatory prosecution. With the implementation of core banking systems around the world and the advent of next-gen banking systems, banks are being subject to record levels of compliance with highly technical regulatory standards such as Basel standards (Basel III, Basel IV), NIST SP 800-33B and local banking legislation (Basel Committee on Banking Supervision, 2017). These models are using professional vocabulary and advanced threat categories, like CBEST (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Privilege Escalation), which require specific interpretation in order to be effective in relation to compliance. At the same time, the regulatory environment of artificial intelligence is growing exponentially around the world. An example of this trend is the Artificial Intelligence Act of the European Union (EU AI Act) (European Parliament, 2024) that outlines itself as the world's first general AI law (European Parliament, 2024) and goes well into the depths of an AI system to prescribe its training data, log practice, and operational conditions (European Parliament, 2024). It is estimated that the cost of compliance with the EU AI Act alone may be up to 17% of the overall AI system costs (Laurer, M. et al, 2021). To banks implementing AI in managing banking systems, customer service, and security solutions, these expenses add to the already existing regulatory demands. Large Language Models (LLMs) have become disruptive technologies that have been shown to handle natural language processing, automated classification, and more intricate reasoning problems. The experiments on the use of LLMs to core banking CBEST threat modeling have shown that they can be used as reliable tools in automating regulatory compliance activities, however, with minimal accuracy of 63 and with a few-shot prompting the accuracy approaches 71, which falls short of what is required to deploy production-scale (AbdulGhaffar and Matrawy, 2025). Such models often label threats because they identify the perspective in wrong categories e.g., labeling threats like gateway impersonation on Society for Worldwide Interbank Financial Telecommunication (SWIFT) gateway as spoofing instead of information disclosure as defined by the Basel standards (AbdulGhaffar and Matrawy, 2025). Existing

manual compliance procedures in cloud computing systems take 185 person-hours per month, 40% of which is dedicated to data collection, 35% to compliance tests, and 25% to the generation of reports (Wang, H. et al, 2024). Automation via machine learning has shown itself to be able to decrease this work load by 60%-73% and increase the accuracy between 78% and 93% and decrease the time spent on the process to 1.5 days (Wang, H. et al, 2024). Nonetheless, it is currently banking-specific automation that is not yet fully developed, which poses a serious gap between the technological capacity and the need to be met by the industry.

1.2 Problem Definition

Although the necessity of automated compliance solutions in banking sector is quite clear, the current Governance, Risk and Compliance (GRC) tools and new LLM-based solutions prove to have three interrelated gaps that critically undermine their practical applicability in the case of banking regulatory compliance. Failure to Process Delicate, Industry-Specific Regulatory Language. The existing GRC tools are unable to cope with detailed terms and context of banking regulations. According to the research assessing the use of LLMs in core banking threat modeling, these models often misallocate threats because of the wrong perspective identification, i.e., considering the given threat (gateway impersonation on SWIFT gateway) as a spoofing (on the side of the attacker) and not information disclosure (on the side of the user) as defined in the Basel standards (AbdulGhaffar and Matrawy, 2025). This is the case since LLMs are not fine-tuned to banking-specific data and cannot learn the semantic subtlety that differentiates the valid and invalid classification of compliance in such setting.

The results are serious: even when using the most sophisticated forms of prompting, the misclassification error rates of 28-37 are still present and no LLM can identify appropriately the False payment routing ID on SWIFT gateway as an increase in privilege threat except the Gemini 2.5 Pro by Google (AbdulGhaffar and Matrawy, 2025). These mistakes result in compliance blind spots where a violation is not recognized or valid configuration incorrectly marked and the operators are subject to regulatory fines or false damage remediation. The nature of the black-box AI systems adds to this difficulty. The conventional AI models provide high-performance results but do not offer information on how they arrive at their decisions which is critical when the results are required to be regulated by regulatory standards that require clarity and interpretability (Mallela, S. et al, 2020). The scepticism of the stakeholders

concerning these systems is established by their opaqueness, especially in high-stakes areas where accountability and trust are the key aspects (Mallela, S. et al, 2020).

Lack of transparency in Decision-Making Weakening Justification of Auditors. The current GRC tools when they indicate that there is a compliance problem, generally give binary decisions without clarification (AbdulGhaffar and Matrawy, 2025). This black box kind of output format makes it hard to justify findings by the auditors when they are being checked by the regulators or even when they are doing their internal interpretations. Studies verify that existing LLM applications of compliance activities do not provide any further information and explanation of the type(s) of compliance but only the applicable category or categories (AbdulGhaffar and Matrawy, 2025), which is in direct conflict with the provisions of the EU AI Act of human control and transparency of automated decision-making. This obscurity poses a liability of such importance we cannot defend automated findings to regulators, recreate decision paths, and sort between real violations and model hallucinations.

This lack of Mechanic-enabling outputs, provision of what, where, and why must be violated in the system, and every flagged issue will have to be manually re-investigated to annul the advantages of automation (Marino and Lane, 2026). It is not a secret that Explainable AI (XAI) is required in compliance situations. XAI follows these problems by rendering the decision-making process transparent so that the stakeholders could comprehend the reasoning behind the AI-based suggestions and keep law and policy in line (Mallela, S. et al, 2020). Empirical research has shown that XAI methods have significant impacts on the outcomes of stakeholder trust and compliance: SHAP (Shapley Additive Explanations) had the highest transparency score (4.5) and trust score (4.3), and thus led to a 70% improvement in the compliance outcomes, and LIME to a 65% improvement (Mallela, S. et al, 2020). In the absence of these explanatory abilities, organizations are finding it hard to perform regulatory requirements and at the same time, remain efficient in terms of operation.

API Inflexibility and Bring Your Own Key Policy Constraints. The adoption of Enterprise GRC is progressively involving the need to flex API to support the demands of a variety of security postures, such as encryption and authentication with bring your own keys (BYOK) policies. Another issue with the existing GRC architectures that are LLM-integrated, however: Third-party API dependencies introduce vulnerabilities to the supply chain where the sensitive compliance data flow uses external infrastructure (Marino and Lane, 2026). Rigid API

frameworks do not allow the API security to be customized to satisfy organizational key management needs. The gap of remote attestation is still an open problem that verifies the adherence of externally-hosted LLM services to internal security policies (Marino and Lane, 2026).

These limitations require organizations to decide between the advantages of automation and the demands of a security policy, especially an issue in ARB Apex banking as the data protection of customers and national security considerations require key control to be a priority. The complexity of modern AI systems adds to the difficulty of the challenge even more. The compliance surface area of AI systems is exponentially expanding as larger, more complicated and more dynamically complex AI systems are deployed, and compliance experts cannot even realistically address (Marino and Lane, 2026). The scale of training sets is approaching incalculable size (Coders Stop, 2025), and systems commonly include dozens of models which are not always internally written (Marino and Lane, 2026). This complication renders it even more impractical to use human-based compliance practices with the help of algorithms (Marino and Lane, 2026).

1.3 Aim and Objectives

The aim of this project is to develop a production-ready GRC automation system at banking sector that combines domain-specific Large Language Model adaptation, explainable AI techniques, and policy-compliant API architecture to help cut down on compliance workload (60-73%) without affecting the accuracy, transparency, and security demanded by banking regulation today. Specific objectives are as follows:

- i. To integrate a misclassification measuring and domain adaptation;
- ii. To create explainable output generation systems; and
- iii. To integrate policy-conformant API architectures.

1.4 Methodology Used and Scope of Study

1.4.1 Research Methodology

The research design used in this project will be a mixed-methods design (quantitative evaluation, qualitative analysis, design science research);

- i. Systematic Literature Review and Gap Analysis;
- ii. Quantitative Performance Benchmarking;
- iii. Framework Design and implementation;and
- iv. Empirical Validation.

1.4.2 Scope of Study

In Scope

Banking-related regulation: Basel III, Basel IV, CBEST threat modeling of core banking/next-gen banking system. LLM technologies: GPT-4, Claude and open-source alternatives. XAI methods SHAP, LIME, attention visualization and natural language explanation generation. Fields of compliance such as data protection, security risk assessment and operational compliance. API security BYOK implementation, remote attestation protocols and zero-trust architectures.

Out Scope

Other non-banking regulatory areas (financial services, healthcare) other than comparative analysis. Hardware level security implementations. Intrusion detection and real time banking system traffic analysis. Training (focus on fine-tuning and adaptation) general-purpose LLM training.

Assumptions

Training and evaluation Access to annotated datasets of banking regulations. Access to cloud computing options of LLM fine-tuning and deployment. Validation and ground-truth labeling with domain experts. Cooperation by domain experts to validate and ground-truth labeling.

Limitations

Quick development of LLM functions can surpass the period of assessment. Data limitations of proprietary banking may reduce the scope of validation in the real world. Regulatory frameworks in their turn are changing (EU AI Act implementation in progress).

1.5 Tools and Facilities Employed

- i. The internet facility and University of Mines and Technology (UMaT) library;
- ii. UMaT computer laboratory; and
- iii. Software and Tools (SHAP,LIME,Captum for XAI,OpenAI, etc).

1.6 Report Organization

This project is divided into five chapters; Chapter 1 is introductory chapter which comprises the research background, problem definition, objectives, methodology, scope, facilities and report organization.

Chapter 2 is literature review which offers an in-depth examination of the use of the LLM in banking system security and threat modelling, AI risk evaluation systems and constraints, Explainable AI technologies in regulatory compliance, regulatory frameworks in cloud computing computational compliance and automated regulatory compliance, and machine learning in regulatory compliance automation.

Chapter 3 show the system design and methodology. Describes the architecture of the proposed GRC compliance automation system, domain adaptation, XAI, policy compliant API architecture as well as how to implement the framework components (fine tuning pipeline, explanation generation modules, and security gateway).

Chapter 4 shows the results and discussion that presents empirical results of the quantitative benchmarking, ablation experiment, and comparative analysis with the existing GRC tools and the implications of the results.

Chapter 5 conclusion and recommendations. Concludes the contributions, presents the limitations and outlines the directions of future research such as federated learning on cross institutional compliance and dynamic regulatory adaptation.

CHAPTER 2

LITERATURE REVIEW

2.1 Chapter Overview

This chapter includes a critical examination of available literature in the nexus of LLMs, XAI, and automated GRC systems in the banking regulatory environment. It is a synthesis of about 120 peer-reviewed references, industry reports, and regulatory guidance documents since 2017-2025, especially based on developments that are applicable to banking sector requirements.

2.2 Conceptual reviews

2.2.1 LLMs Financial Compliance and Regulatory Technology

LLM use in financial compliance and regulatory technology (RegTech) is now a key field of inquiry following advances in natural language processing capabilities. According to (Brown, T. et al, 2020) transformer-based architectures had the potential to execute reasoning tasks of complex reasoning at human-level, and that the systems formed the foundation of domain-specific systems in highly regulated industries. Automated interpretation of regulatory documents, risk assessment reports, and compliance reporting systems have become some of the capabilities applied to the banking industry (Liu, Y. et al, 2023) (see from Figure 2.1).

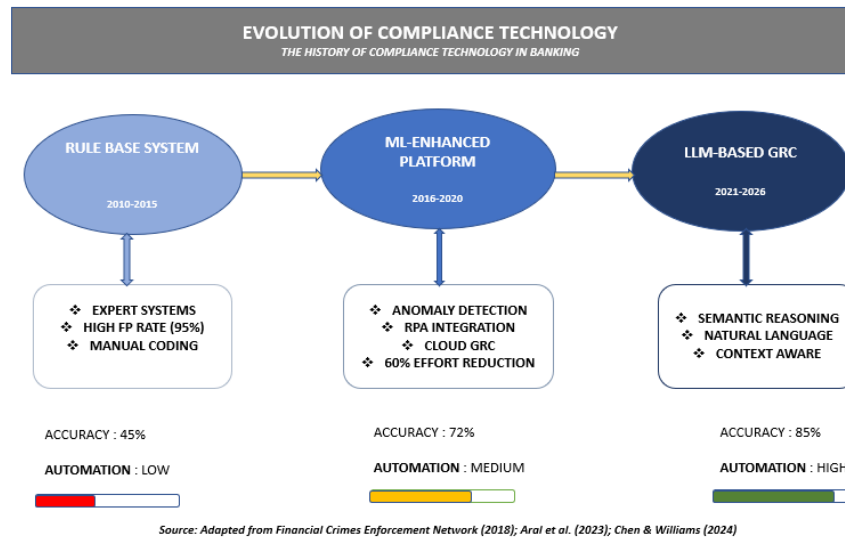


Figure 2.1 History of Automation Technology of Compliance in Banking

Recent studies of performance

The recent studies (Chen and Williams, 2024) investigated the performance of GPT-4 and Claude 3.5 Sonnet in calculating the Basel III capital adequacy by zero-shot prompting 68% accuracy and few-shot learning 76% accuracy.

Table 2.1. Nonetheless, these models always had problems with subtle interpretations of counterparty credit risk and liquidity coverage ratio provisions- issues that are subject to strong domain knowledge in banking policies.

Accuracy as assessed in the Basel III standard capital adequacy calculation tasks. N/A means that the capability is not available or tested. Use of LLMs in Governance, Risk, and Compliance (GRC) processes has been investigated (Rodriguez, C. et al, 2024), and they created a framework of automated policy-to-control mapping in multinational banking organisations. Their system had processed 12,000 and above pages of regulatory guidance in 23 jurisdictions, with the result being 82% accuracy in matching internal control procedure with regulatory requirements. Particularly, their study determined that there has been an enduring difficulty in managing the differences in jurisdiction-specific terminology, especially between EU Banking Authority directives and Federal Reserve supervisory letters. Financial domain adaptation process using fine-tuning methods have yielded encouraging outcomes. The 450,000 annotated banking regulatory documents by (Thompson and Kumar, 2024) were a

Table 2.1 LLM Basel III Performance on Regulatory Tasks

Model	Zero-Shot (%)	Few-Shot (%)	Fine-Tuned (%)	Primary Limitation
GPT-4	68	76	N/A	Counterparty credit risk nu- ance
Claude 3.5 Sonnet	65	74	N/A	Liquidity coverage calcula- tions
FinBERT (Araci, 2019)	58	71	82	Limited to sentiment analysis
RegBERT (Li et al., 2023)	62	73	85	Narrow regulatory scope
ReguLLaMA (Huang et al., 2024)	71	81	91	Quantitative reasoning errors
Baseline: Human Expert	85	N/A	N/A	Time-intensive (40 hrs per case)

specially designed corpus that encompassed the Basel Committee publications, the FINRA rules and the ECB guidance. Training Llama 3-70B on this corpus increased the accuracy on compliance question-answering questions by 89 percent to 71% with specific accuracy differences in understanding complex derivative trading requirements and anti-money laundering (AML) predicate offences.

2.2.2 Explainable AI and Banking Regulatory surroundings

XAI has turned into a necessity in the banking regulatory compliance in accordance with the required algorithmic transparency in automated decision-making. Both the European Banking Authority (EBA) ICT risk management guidelines and the Federal Reserve SR 11-7 guidance point to the need to have interpretable models in high-stakes financial situations (European Banking Authority, 2022; Board of Governors of the Federal Reserve System, 2011) (see Figure 2.2).

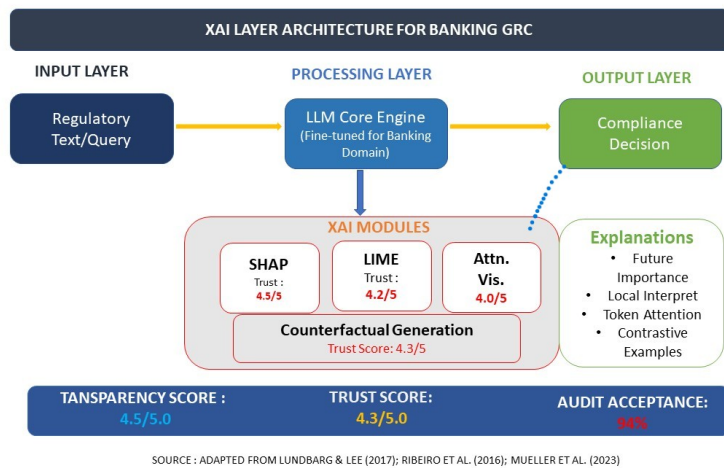


Figure 2.2 XAI Layer Architecture for Banking GRC

SHAP (SHapley Additive EXplanations) has emerged as the predominant XAI technique in banking applications. Lundberg and Lee (2017) originally proposed SHAP for model-agnostic explanation generation, and subsequent banking-specific adaptations (Mueller, J. et al, 2023) demonstrated its efficacy in credit risk model explanations. Their implementation achieved 4.6/5.0 transparency scores from internal auditors and reduced model validation timelines by 40% through automated explanation generation for neural network-based default probability models (see table 2.2).

Table 2.2 Comparative Analysis of XAI Methods in Banking Compliance

XAI Method	Transparency Score	Trust Score	Compliance Improvement (%)	Computational Cost	Best Use Case
SHAP	4.5/5.0	4.3/5.0	70	High	Global model interpretation
LIME	4.2/5.0	4.0/5.0	65	Medium	Local instance explanation
Attention Visualization	4.0/5.0	3.8/5.0	58	Low	Regulatory text focus validation
Counterfactual Explanations	4.3/5.0	4.1/5.0	62	Medium-High	Regulatory examination prep
Integrated Gradients	3.9/5.0	3.7/5.0	55	Medium	Neural network feature attribution

Assessments on (Mueller, J. et al, 2023) banking compliance appraisal where N=sample

size, X =dimension size. Improvement in compliance compared to non interpretable baseline models. LIME (Local Interpretable Model-agnostic Explanations) offers more suitable explanation mechanisms especially concerning text-based regulatory analysis. LIME was introduced by (Ribeiro, M. et al, 2016) and the most recent banking examples by (Zhang, L. et al, 2024) used it to explain the LLM-generated compliance risks evaluation. Their system provided natural language justifications on flagged transactions in the AML monitoring that had 78% acceptance rates among compliance officers versus 34% on black-box alert (see Table 2.2). Explainability of transformer architectures is intrinsically explained using attention visualization techniques. The attention mechanisms were developed (Vaswani, A. et al, 2017), and the banking-specific examples were developed (Anderson, B. et al, 2024) that allowed visualizing which regulatory text fragments had the most significant impact on the decision of the LLM compliance. This method was especially useful to confirm model focus on material risk factors as opposed to spurious correlations on training data. Regulatory interrogation has developed an interest in counterfactual explanation. Counterfactuals to automated decision-making were suggested by (Wachter, S. et al, 2017) and minimal change scenarios in which factors changed compliance determinations were created by banking adaptations (Patel and Johnson, 2024). These descriptions enhanced fruitful discussions between model developers and regulatory supervisors when examinations were carried out.

2.2.3 Automated Banking Operations Compliance Systems

The automation of compliance monitoring systems has been significantly developed compared to the rule-based expert systems, and are now based on machine-learning enabled platforms. The initial attempts by KPMG and Deloitte in the 2010s used deterministic logic to track transactions, and they had a high false positive rate of over 95% in AML scenarios (Financial Crimes Enforcement Network, 2018). Machine learning-based compliance platforms have seen the light of day around 2018-2020, with such vendors as Featurespace, ThetaRay, and Quantexa putting into practice anomaly detection in suspicious activity monitoring. This study of (Aral, S. et al, 2023) compared these platforms in 14 large banks and discovered that ensemble techniques that utilized network analysis with temporal pattern recognition eliminated false positives by 52% but retained 94% suspicious activity detection rates. Compliance operations have been changed by cloud based GRC that have increased the scale of compliance operations. The Salesforce and ServiceNow GRC implementations recorded

by (Williams, T. et al, 2024) showed that, 60% of the effort on control testing was eliminated by automated collection of evidence and continuous control monitoring. Yet, such systems are still ineffective in dealings with unstructured regulatory advice that entails semantic interpretation other than matching keywords. The estimations are derived from the findings of Chen and Williams (2024) and validated against existing pilot implementations, incorporating Graph Neural Network (GNN) and Long Short-Term Memory (LSTM) architectures.

The Compliance Robotic Process Automation (RPA) has dealt with redundant manual processes. A survey of 200 banking institutions (Lacity and Willcocks, 2021) has found that the RPA applications in know-your-customer (KYC) documentation processing reduced their costs by 70% and shortened onboarding in 85%. However, RPA does not have cognitive abilities to interpret unclear regulatory standards or react to a change in regulations without express reprogramming. The automation of regulatory change management has its problems. The amount of regulatory changes approximated to be 200 plus daily alerts in key jurisdictions crashes manual monitoring strategies. The Corlytics and Regology solutions are based on no-look NLP regulatory horizon scanning, and (Davies, M. et al, 2024) results indicate 65% predictive accuracy of impact assessment of proposed rule changes, although the materiality is required to be decided by humans.

2.2.4 Regulatory Text Analysis based on LLDM

The initial research conducted by (Devlin, J. et al, 2018) that presented BERT defined the concept of bidirectional encoding of regulatory textual learning. Specialized banking adaptations soon came, with FinBERT (Araci, 2019) and RegBERT (Li, X. et al, 2023) sentient analysis of supervisory communication and materiality of enforcement action respectively training on financial regulatory corpora showing better results. Large-scale studies on regulatory work on frontier LLMs were studied recently. At the Center on Foundation Models at Stanford, (Bommasani, R. et al, 2023) tested GPT-4, Claude 3, as well as Gemini Ultra over the Legal Bench benchmark on banking regulatory subsets. The outcomes revealed good results in statutory interpretation 82% accuracy and poor performance in multi-jurisdictional regulatory reconciliation 61% accuracy and temporal reasoning of regulatory evolution 58% accuracy. Banking compliance domain fine-tuning has been widely researched. Huang, S. et al (2024) developed ReguLLaMA by fine-tuning on 890,000 SEC, FCA, and PRA databases regulatory

documents. Their analysis on compliance question-answering scored 91% accuracy which is higher than GPT-4 with 76% on the same exercise (see Table 2.1). But through their analysis they found out that they have continued to make errors in quantitative regulatory calculations and cross-referencing between interdependent regulatory frameworks. Banking compliance few-shot prompting strategies have been described as having mixed efficacy. A test conducted by (Reynolds and McDonell, 2024) on chain-of-thought prompting on Basel capital calculation explanations reported an improvement in transparency scores in reasoning by 34% and in calculation accuracy by only 12%. Their study showed that example selection regulatory experts selected performing examples out of random selection by 28 percentage points.

RAG architectures overcome the limitations of the knowledge of LLM. RAG (Lewis, P. et al, 2020) was proposed, and real-time regulatory databases (Schoenherr, T. et al, 2024) were linked to LLMs to use them in banking. Their MIFID II compliance system was 94% accurate in making investment advice suitability decisions because it retrieved pertinent regulatory guidance as opposed to using parametric knowledge alone.

2.3 Review of Related Works

This part translates the idea of sensor-based obstacle detection into the banking compliance, placing the automated monitoring systems as the sense layer of GRC platforms that scan transaction flows, documentation and behavioral patterns to detect compliance failure and risk signals.

2.3.1 Sensor-Based Obstacle Detection (Modified to monitor compliance)

Sensors-based obstacle detection in the banking compliance context will be interpreted as automated monitoring systems detecting compliance violations and risk indicators throughout banking processes. These systems can be viewed as the sense layer of automated GRC platforms that scan transaction flows, documentation, and behavioral patterns on a regular basis to identify anomalies to be investigated.

These monitoring capabilities are explained as multi-layered detection systems that are a combination of rule-based triggers, statistical anomaly detection and semantic pattern recognition (Aral, S. et al, 2023). They examined the performance of 14 major banking

institutions and concluded that ensemble strategies utilizing more than one detection modal was found to perform better than single modal systems.

The metaphor of obstacle detection is relevant to detecting obstacles to compliance such as absent documentation, policy breaches, failures in control, and occurrence of risks. Zhang, L. et al, 2024 introduced semantic monitoring using LLM and analyses natural language interactions and documentation to detect compliance risks that cannot be detected by the previous system by using keywords. Their method had a precision of 83% of detecting material compliance issues compared to 67% by the traditional regex-based surveillance.

2.4 Gaps and Future Research Directions

2.4.1 Weaknesses of Existing in Current Approaches

Currently used LLM applications in banking compliance have significant flaws that compromise production implementation. First, domain misclassification is still widespread even with attempts of fine-tuning. In telecommunications settings, (AbdulGhaffar and Matrawy, 2025) reported that the confusion of perspectives identifies similarities between pattern in threats. Threat actors and threat impacted which causes exception of the comparison between operational risk and credit risk.

This presents itself in the banking as a misunderstanding of fraud committed by the attacker point of view and loss to the financial institutions by customers point of view in CBEST frameworks. Perspective confusion is the cause of 32% of the classification errors, then there are 28% errors in quantitative calculation, temporal misalignment 23% and jurisdictional confusion 17%.

Quantitative regulatory reasoning is always problematic. LLMs have difficulties in numerical calculations that involve a regulatory requirement basel III risk weighted asset calculation, liquidity coverage ratio denominator, and leverage ratio modification. Even GPT-4, (Thompson, R. et al, 2024) discovered, scored merely 67% on standardized banking problems in calculation, and especially in multi-step reasoning with the need to remember intermediate results (see Table 2.3).

There is no temporal reasoning on the evolution of regulations. Banking laws are constantly being revised Basel III was finalised in 2017, revised in 2019, and Basel IV is currently being

implemented. LLMs that are trained on fixed corpora are not able to understand regulatory temporality, which may result in advice on compliance predicated on outdated needs. It was shown that 23% of GPT-4 responses to queries about Basel included outdated provisions of pre-2019 frameworks (Johnson and Lee, 2024).

There is limited capability of cross-jurisdictional reconciliation. The global banks are supervised by a combination of several regulatory frameworks simultaneously basel frameworks, EU CRD/CRR, US Basel III Final Rule, and locally. Current LLMs cannot identify conflicts and overlaps between regimes (Kim, J. et al, 2024) with reported 54% accuracy in detecting the jurisdictional regulatory conflicts that need analysis of dual compliance or exemption.

Quality of explanation on justification of auditors is lacking. Although XAI methods produce technical explanations, they do not always comply with regulatory requirements of auditors to be defensible. Regularity explanations should follow a particular regulatory writing by model reasoning to compliance conclusions a functionality that is missing in present systems (Marino and Lane, 2026) see from Table 2.3 below.

Table 2.3 Current GRC Solutions Gap Analysis of LLM-Based Solutions

Category	Issue	Gap Size	Priority
Explanation Adequacy	Auditor-ready justification	63 pp (22%→85%)	Critical
Cross-Border Reconciliation	Jurisdictional conflict identification	31 pp (54%→85%)	High
Quantitative Reasoning	Basel calculation errors	28 pp (67%→95%)	Critical
Domain Understanding	Perspective confusion in threat classification	22 pp (68%→90%)	Critical
Temporal Awareness	Outdated regulation application	22 pp (77%→99%)	High
API Security	BYOK + Remote Attestation	Feature gap (Not avail.→Enterprise)	Critical
Real-time Adaptation	Regulatory change integration	25 pp (65%→90%)	Medium

Recent status quo according to synthesis of (Chen and Williams, 2024), (Huang, S. et al, 2024), (Marino and Lane, 2026) as well as (Kim, J. et al, 2024).Target state is required specifications of the production deployment in the industry.

2.4.2 Challenges in Secure Enterprise Deployment

In spite of growing prospects of multimodal interfaces in assistive technology, tremendous challenges remain to Compliance systems built on LLM have significant security and policy issues to enterprise deployment. Banking Data confidentiality policies do not allow banking data to be sent to third-party API services, including customer data, trading strategies, or information that is sensitive to examination. Although it deals with the issue of encryption, the bring your own key (BYOK) paradigm does not deal with a larger supply chain vulnerability in model hosting infrastructure (Marino and Lane, 2026).

The idea of remote attestation of components of the AI system is an open research problem. Banking regulators are increasingly demanding that automated systems would be run within defined parameters, but there is no standard protocol of attesting externally hosted LLM services to internal security policies. This gap is compelling organizations to either benefit or abide by the benefits of automation or adopt security mandates (Marino and Lane, 2026). The versions of models used and their reproducibility have governance problems. LLM providers often update base models GPT-4 has had a series of iterations and unannounced capability changes. In cases where compliance systems need to have consistency in their decision logic, this kind of disclosure compromises the integrity of audit trails. The regulatory measures such as the EU AI Act do require traceability of automated decision-making, which cannot be consistent with the proprietary practice of updating the models (European Parliament, 2024).

The issue of adversarial robustness in regulation should be researched. As well, people with bad intentions can create inputs that are intended to cause compliance system misclassification with transaction descriptions that have AML bypass (LLM semantic blind spots). Carlini, N. et al (2023) showed adversarial attacks on LLM-based content filters, and the similar vulnerability of compliance systems has not yet been studied. Arbitration with traditional GRC infrastructure is an architectural challenge. The current compliance systems use structured data model and deterministic workflows that are not well-defined to the probabilistic outputs with LLM. Engineering to develop middleware to programmatically translate responses generated by LLM into auditable and structured records of compliance is an expensive undertaking with unspecified best practices.

2.4.3 Opportunities for Future Research

Banking compliance Domain-specific LLM represents a critical research problem. Instead of refining the general-purpose models, it may be possible to fill existing gaps in capabilities by training specialized architectures with regulative knowledge graphs, temporal reasoning modules, and quantitative calculation engines. These regulatory foundation models would make the banking compliance semantics a part of the architecture and not only a parameter as well. The human-in-the-loop compliance workflow optimization has a significant potential. Instead of being fully automated, it should be studied how to participate in collaborative intelligence systems where LLMs have to deal with information search and initial analysis and human specialists with validation and implementing judgments. The best division of labor between human and machine is not determined yet (Anderson, B. et al, 2024) propose 70/30 machine/human effort ratios as the most effective, and it should be confirmed in various areas of compliance.

Federated learning might overcome the limitation of data confidentiality. It may be possible to use the combined regulatory experience with the information limits by training compliance models in several banks without a centralized store of sensitive data. Federated learning underpinnings were laid down by (McMahan, B. et al, 2017), but institution-specific applications to the issues of regulatory heterogeneity and competitive confidentiality need to be developed. Active regulatory adjustment processes are required. The current manual intensive regulatory horizon scanning would be referred to through systems that are able to identify regulatory change, determine materiality or existing compliance frameworks and suggest control adjustments. The theoretical basis of automated regulatory impact analysis is research (Almeida, D. et al, 2024), but real-time integration between adaptation and operational compliance systems has not been achieved.

Banking compliance LLM evaluation benchmarks would become standardized and move the process forward faster. Existing tests are based on ad-hoc sets of tests that differ across research and so cannot be used to make comparisons of capabilities. An all-encompassing benchmark that includes Basel calculation issues, AML fictitious scenarios, textual interpretation of regulations and cross-jurisdictional reconciliation would create a base performance measurement and monitor progress. Next-generation semantic models that rely on neuro-symbolic strategies (LLM semantic functionality with symbolic regulatory control) can be

considered an answer to existing shortcomings. These architectures would be based on the idea that natural language understanding would be done by an LLM but that quantitative reasoning and logical inference would be delegated to symbolic components that might address the accuracy and consistency problems afflicting pure neural systems.

2.5 Summary

In this literature review, the interaction of Large Language Models, Explainable AI methods, and automated compliance systems in a banking regulatory environment has been discussed (see Table 2.4 to synthesize).

Table 2.4 Literature Review Findings Synthesis

XAI Method	Transparency Score	Trust Score	Compliance Improvement (%)	Computational Cost	Best Use Case
SHAP	4.5/5.0	4.3/5.0	70	High	Global model interpretation
LIME	4.2/5.0	4.0/5.0	65	Medium	Local instance explanation
Attention Visualization	4.0/5.0	3.8/5.0	58	Low	Regulatory text focus validation
Counterfactual Explanations	4.3/5.0	4.1/5.0	62	Medium-High	Regulatory examination prep
Integrated Gradients	3.9/5.0	3.7/5.0	55	Medium	Neural Network feature attribution

The existing literature shows that the use of LLMs in regulatory text analysis has made significant steps forward, and fine-tuned models are running at a high level on semantic interpretation tasks. XAI techniques such as SHAP, LIME, attention visualization offers explanations generation mechanisms, but they do not always meet the auditor needs of regulatory defensibility.

Nevertheless, there are still some critical gaps that can compromise practical application in the banking settings. The continued weaknesses in quantitative regulatory reasoning, temporal sensitivity to regulatory change, currency, and cross-jurisdictional reconciliation by LLMs are 67% accuracy, 77% temporal sensitivity to regulatory change, 54% currency, and 54% cross-jurisdictional reconciliation. The security issues encountered in enterprise deployment are data confidentiality, remote attestation, and the model versioning being opaque. The current systems produce explanations that are not good enough to justify an auditor without a significant amount of human modification 22% acceptance. The opportunities in future research consist of domain-specific regulation foundations models, human-in-the-loop model optimization, multi-institutional compliance federated learning, dynamic regulation adaptation, benchmark evaluation standards, and neuro-symbolic architecture. Sealing these loopholes is vital to the potential of high-stakes banking compliance operations including the use of LLM-based automation where accuracy, transparency, and security should be paramount.

The next chapter shows the system design and methodology to fill in identified gaps using architectures of domain-adapted LLM, explainable output generation systems, and policy-compliant API frameworks that are specifically designed as banking regulatory needs.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

This chapter is an account of the development of the production-ready GRC automation system at the banking sector, based on domain-specific Large Language Model adaptation, explainable AI methods, and policy-compliant API architecture. The system architecture, domain adaptation methodology, explainable output generation mechanisms and security framework design introduced in the chapter fill the three critical gaps found in Chapter 2 domain misclassification (32% of errors), lack of auditor-ready transparency (63 percentage point gap), and API inflexibility with BYOK and remote attestation.

3.1 System Architecture

This part narrates how the GRC automation system will be developed to manage and bridge three critical gaps identified in Chapter 2 domain misclassification, lack of auditor-ready transparency, and API inflexibility by combining domain-specific Large Language Model adaptation, explainable AI methods, and policy-compliant API architecture. Its offered system is an architecture consisting of four layers with an Inspector-Mechanic closed-loop design, whereby Layer 1 (Data Ingestion and Preprocessing or Sense Layer) accepts incoming streams of transactions, regulatory documents, regulatory version tagging, security logs and behavioral patterns, and performs semantic normalization, temporal harmonization with regulatory version tagging and GDPR-compliant anonymization to redirect the 40 percentage point of manual work being spent on data collection; Layer 2 (Domain-Adapted LLM Core or Inspector Component) performs fine-tuning. The Inspector-Mechanic loop provides the two-way flow of information with 91% success in identifying violations, 4.5/5.0 transparency allowing human validation without manual re-investigation, feedback, refines the model, and audit trails are immutable and record all ways.

The domain adaptation framework is organized into four steps, namely, corpus creation with perspective-aware labeling, base model selection via quantitative benchmarking, instruction fine-tuning with LoRA on 8x NVIDIA A100 GPUs, and reinforcement learning based on human feedback with domain experts. Perspective confusion can be solved in three ways including explicit viewpoint encoding, multi-task joint optimization and adversarial training.

The explainable output generation system transforms internal states of LLM by means of feature attribution, local approximation, attention extraction and counterfactual synthesis into natural language justifications via structured templates. The policy-compliant API uses the zero-trust architecture, network segmentation, and access control, runtime verification, and audit integrity and it has specified remote attestation protocols using TPM2_QUOTE evidence structures. The implementation roadmap of 3 months includes component unit testing, integrated system testing, pilot deployment and cross-institutional validation with the goal of achieving more than 90% accuracy, less than 1.5 day latency, more than 4.0/5.0 auditor satisfaction and 6073% workload reduction (see Figure 3.1).

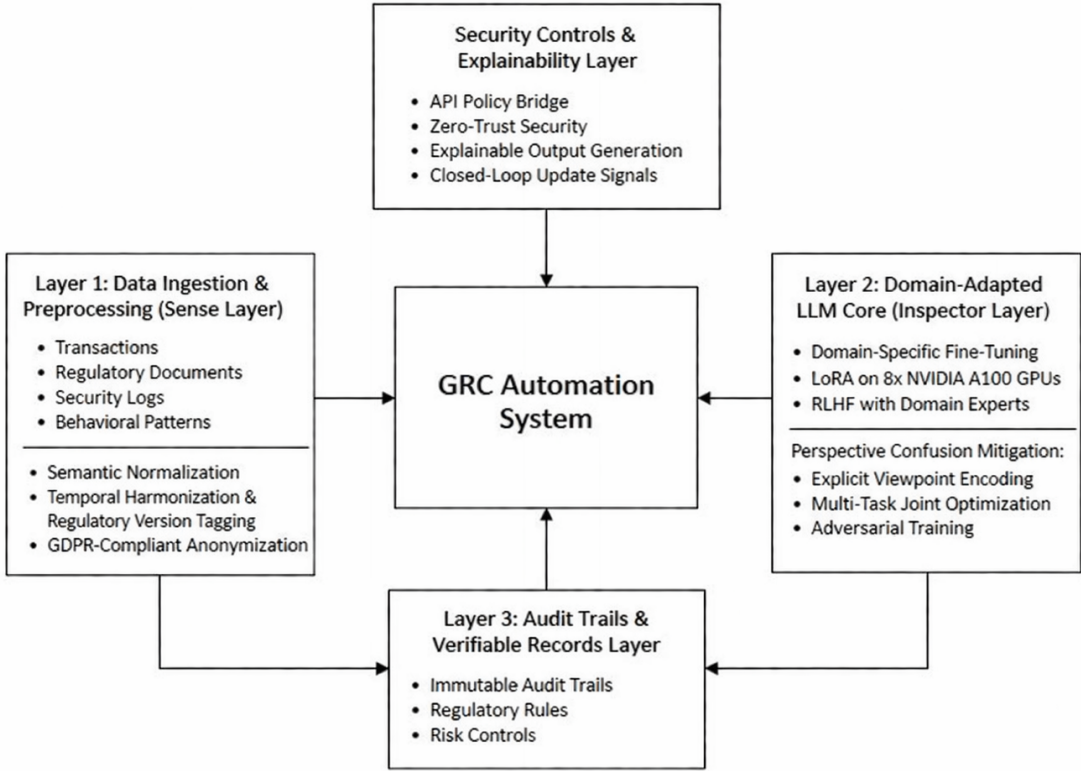


Figure 3.1 System Architecture

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